

A1.5: Simulation

1. Why Simulate?

Simulation models real-world processes using random sampling, when analytical solutions are intractable or when the system involves randomness (queues, arrivals, breakdowns).

2. Random Numbers

Computers generate **pseudo-random** numbers $U \sim U(0, 1)$. These drive the simulation via transformations.

Inverse Transform Method

If F is the CDF of target distribution and $U \sim U(0, 1)$, then $X = F^{-1}(U)$ has CDF F .

Simulate $\text{Exp}(\lambda)$.

$$F(x) = 1 - e^{-\lambda x}. X = -\ln(U)/\lambda.$$

Simulate Bernoulli(p).

Return 1 if $U < p$, else 0.

Simulate discrete distribution (e.g., dice roll).

$$X = \lceil 6U \rceil \text{ for fair die.}$$

3. Monte Carlo Methods

Estimate a quantity by repeated random sampling. Error scales as $O(1/\sqrt{n})$.

Estimate π by throwing darts.

Generate $(U_1, U_2) \in [0, 1]^2$. Check $U_1^2 + U_2^2 < 1$. Ratio $\times 4 \approx \pi$.

4. Event-Driven Simulation (Queues)

Model queue via two event types:

- **Arrival:** customer enters at time t , sampled from inter-arrival distribution.
- **Departure:** server finishes at time t , sampled from service-time distribution.

Advance simulation clock to next event, update queue state, record statistics, repeat.

Little's Law

$$L = \lambda W$$

Mean number in system = arrival rate $\times$ mean time in system.

M/M/1 Queue

Poisson arrivals rate λ , exponential service rate μ . Utilisation $\rho = \lambda/\mu$. Steady-state (if $\rho < 1$):

$$L = \frac{\rho}{1 - \rho}, \quad W = \frac{1}{\mu - \lambda}$$

5. Variance Reduction

- **Antithetic variates:** use pairs $(U, 1 - U)$ to reduce estimator variance.
- **Common random numbers:** reuse same draws across scenarios.
- **Control variates:** adjust using correlated variable with known mean.

6. Validation and Verification

- **Verification:** does the code do what it should?
- **Validation:** does the model reflect reality?
- **Warm-up:** discard initial transient before measuring steady-state.

7. Exam Strategy

- State clearly the random variable distribution and how to sample from it via $U \sim U(0, 1)$.
- For Monte Carlo estimation, include approximate standard error or confidence interval.
- Queue simulations: tabulate event times and queue state step by step.